

Stable marriage (matching) problem Gale-Shapely algorithm

A boy for a girl or a girl for a boy is called *possible* if they are a couple in at least one stable matching.

A stable matching M is called *boy-optimal* if for each of the boys their pair in M is the best possible pair for him.

A stable matching M is called *girl-pessimal* if for each of the girls their pair in M is the worst possible pair for her.

1. Run Gale-Shapely on the following instance

$b_1 : (g_3, g_2, g_5, g_1, g_4)$

$b_2 : (g_1, g_2, g_5, g_3, g_4)$

$b_3 : (g_4, g_3, g_2, g_1, g_5)$

$b_4 : (g_1, g_3, g_4, g_2, g_5)$

$b_5 : (g_1, g_2, g_4, g_5, g_3)$

$g_1 : (b_3, b_5, b_2, b_1, b_4)$

$g_2 : (b_5, b_2, b_1, b_4, b_3)$

$g_3 : (b_4, b_3, b_5, b_1, b_2)$

$g_4 : (b_1, b_2, b_3, b_4, b_5)$

$g_5 : (b_2, b_3, b_4, b_1, b_5)$

2. (a) Show, that there at most $n(n-1) + 1$ proposals in the GS algorithm.
(b) Show, that the algorithm terminates in at most $(n-1)^2 + 1$ days.
3. Show, that GS is boy-optimal.
4. Show, that GS is girl-pessimal.
5. Show a stable matching which is neither boy- nor girl-optimal.